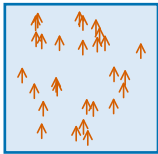
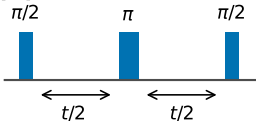
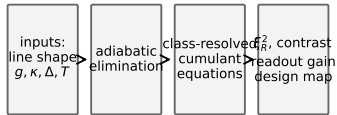


(a) $^{171}\text{Yb}^{3+}:\text{CaWO}_4$ cavity κ, Δ  $\gamma_{\text{inh}}, T_2, n_{\text{th}}, g_j$ **(b)**

$$H_{\text{eff}} = \chi \hat{J}_+ \hat{J}_- + \sum_j \frac{\delta_j}{2} \hat{\sigma}_j^z$$

$$\hat{L}_\downarrow = \sqrt{\Gamma_{\text{SR}}(n_{\text{th}} + 1)} \hat{J}_-, \quad \hat{L}_\uparrow = \sqrt{\Gamma_{\text{SR}} n_{\text{th}}} \hat{J}_+$$

echo twist: OAT survives, static disorder refocused

(c)

validated against exact master equations
and the analytic $(\Gamma_{\text{SR}}/\chi)^{2/3}$ law